

AWADHOOT LOHARKAR

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 github.com/AwadhootLoharkar

Education

QuanTEEM — Erasmus Mundus Joint Master

Sep 2024 – Sept 2026

M.Sc. in Quantum Technologies and Engineering

France, Germany

- Sem 1 (Université de Bourgogne): **15.59/20** — Sem 2 & 3 (RPTU Kaiserslautern): **1.7**
- Erasmus Mundus Scholarship holder (~4% acceptance rate).
- Relevant Coursework: Quantum Optics , Quantum Field Theory , Quantum Mechanics, Quantum Algorithms, Quantum Control, Solid-State Physics, Numerical Methods, Quantum Computing in AI.

Fergusson College

Sept 2020 – April 2023

Bachelor of Science (Physics)

Pune, India

- Grade: 9.00/10 CGPA. Thesis: Analysis of QKD protocols for Post-Quantum Cryptographic frameworks.

Experience

Master Thesis

March 2026 – Sept 2026

Fraunhofer ITWM — QuEnAIS Project

Kaiserslautern, Germany

- Engineered a hybrid DMET framework for quantum simulations of strongly correlated protein-ligand systems, mapping high-dimensional active spaces onto optimized quantum circuits.
- Deploying and optimizing advanced quantum impurity solvers (SQD, SKQD, and SqDRIFT) to accurately resolve complex electron correlation in pharmacologically relevant molecules
- Constructing self-consistent DMET loops to extract precise quantum descriptors (HOMO/LUMO, Fukui indices).
- Developed **QuEnAIS**, an open-source software package implementing the simulation pipeline (in active development): github.com/AwadhootVijayLoharkar/quenais

Semester Research Student

Oct 2025 – Feb 2026

Sebastian Eggert Group, RPTU

Kaiserslautern, Germany

- Researched on ground-state of the 1D extended anyonic Hubbard model using DMRG, VQE, and SKQD.
- Obtained consistent results in the hard-core boson limit with results depending on non commuting terms.
- Demonstrated superior convergence of SKQD compared to VQE, achieving 0.05% vs. 10% relative error with respect to DMRG for 5 to 15 site lattices

Research Assistant (HiWi)

Jun 2025 – Feb 2026

Fraunhofer ITWM

Kaiserslautern, Germany

- Project 2 : Researched scalable multi-node and multi-GPU tensor-network simulations for quantum many-body physics and quantum chemistry using QUIMB, ITensors, Qiskit and cuQuantum for HPC Clusters.
- Developed a unified Python toolkit and benchmarked accuracy, scaling, and multi-GPU performance on custom Hamiltonians.
- Project 1 : Worked on the DLR-funded QUANT²AI project, evaluating end-to-end QML pipelines on classical HPC and quantum hardware.
- Developed benchmarking tools comparing performance, scalability, and practical suitability across backends.

Quantum Research Intern

Dec 2023 – Sep 2024

Creed & Bear

Remote

- Contributed to hybrid quantum-classical optimization for power distribution networks (TCS Quantum Challenge 2023); Investigated post-quantum cryptography vulnerabilities.

Quantum Research Intern — QWorld Research Internship Programme

Aug 2023 – Dec 2023

Classiq Technologies

Remote

- Implemented quantum oracle construction and amplitude amplification circuits using the Classiq synthesis platform, gaining experience with high-level-to-gate-level compilation and circuit depth optimisation for near-term hardware.

Quantum Intern

May 2023 – Aug 2023

Silicofeller Quantum

Delhi, India

- Got introduced and worked with Qiskit Metal to design custom 5 qubit superconducting circuit layouts for application-specific quantum processors, developed familiarity with transmon-based device geometry, coupling structures.

Preprints and Manuscripts

- *Assessing Algorithmic Robustness of SKQD and VQE in Non-Commuting Many-Body Hamiltonians* — Preprint. Available at: <https://doi.org/10.13140/RG.2.2.21386.63683>
- *Performance Analysis of Tensor Network Libraries for GPU-Based Quantum Simulation of One-Dimensional Extended Hubbard Model* — manuscript in preparation.

Skills

- **Programming Languages:** Python, Julia, C, MATLAB, Bash.
- **Quantum Packages:** Qiskit, Cuda-Q, Classiq, PennyLane, qBraid, CuQuantum.
- **Tools:** TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib, Git, Linux, MPI, PySCF, ITensors, Block2, QUIMB.

Projects

Iterative Hamiltonian Mapping in SqDRIFT Algorithm

May 2026

Python, Qiskit, NumPy, Quantum Simulation

- Modifying the SqDRIFT algorithm to support iterative addition of Hamiltonian terms, optimizing the decomposition of complex operators for real-time quantum dynamics simulations.
- Developing a custom sampling logic that dynamically updates the drift coefficient, reducing the gate overhead required to reach target precision compared to standard randomized Trotterization.
- Simulating the modified algorithm on multi-qubit systems on real Superconducting hardware.

Referees

- **Dr. Arcesio Castaneda Medina** — Senior Researcher, Fraunhofer ITWM
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- **Prof. Sebastian Eggert** — Professor of Theoretical Physics, RPTU Kaiserslautern
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